Study Questions: Agents

1. Describe a moment in your own crochet practice when you experienced a "flow state" — everything felt smooth and predictable. How does this relate to the concept of low free energy?

2. When you pick up a pattern you have never tried before, your generative model is sparse. How does the experience of following new instructions differ from working a familiar pattern, and what does this tell you about generative model depth?

3. Compare the decision to frog an entire piece versus accepting a small error and continuing. What factors in the Active Inference framework might determine which strategy the crocheter-as-agent chooses?

4. A crocheter counts their stitches at the end of every row. In Active Inference terms, what kind of process is this, and why is it important for maintaining low free energy?

5. Experienced crocheters often say they can "feel" when they have made a mistake before they see it. What does this suggest about the precision of their generative model compared to a beginner's?

6. You are crocheting a hat and the crown shaping starts to pucker instead of lying flat. Describe this situation in terms of prediction error: what did you expect, what did you observe, and what options do you have as an agent?

7. How does your generative model of yarn behavior differ between yarns you have used many times (like a favorite worsted acrylic) and a fiber you have never tried (like bamboo or silk)?

8. A crocheter decides to intentionally deviate from a pattern, adding extra rows to make a longer scarf. Is this a failure of the generative model, or is it an example of the agent updating the model? Explain.

9. When you teach someone to crochet, you are essentially helping them build a generative model. What are the first elements of the model that a brand-new crocheter needs to acquire?

10. In Active Inference, agents minimize free energy either by changing the world or changing their model. Give one crochet example of each strategy.

11. Consider the concept of "gauge" — the number of stitches and rows per inch. How does making a gauge swatch function as a test of the crocheter's generative model against reality?

12. A crocheter who has been working for two hours realizes they have been crocheting with the wrong color yarn. Describe the free energy spike this produces and the agent's likely response.

13. Some crocheters prefer to work without a written pattern, designing as they go. In Active Inference terms, how does the generative model of a freeform crocheter differ from one who follows a pattern?

14. When you frog a row and redo it, the second attempt usually goes faster and more smoothly. Why? What has changed in your generative model between the first and second attempt?

15. How does the size and complexity of a project affect the demands on the crocheter-as-agent? Compare making a simple dishcloth to making a lace tablecloth in terms of generative model complexity.

16. A crocheter joins a crochet circle and sees someone else's beautiful finished project in a stitch they have never tried. Describe the motivational process in Active Inference terms — what drives the agent to attempt something new?

17. The concept of "expected free energy" involves evaluating future outcomes. How does a crocheter use expected free energy when deciding whether to attempt a challenging pattern or stick with a familiar one?

18. When you switch from one stitch type to another mid-project (e.g., from single crochet to shell stitch), your generative model must shift. Describe the brief transition period and the prediction errors that might arise.

19. An advanced crocheter can look at a finished garment and reverse-engineer the construction. How does this ability reflect a deep generative model? What can the advanced agent "see" that a beginner cannot?

20. Reflect on a time you had to frog a significant amount of work. Describe the emotional and cognitive experience through the lens of Active Inference: what was the prediction error, what was the cost of action, and what did you learn that updated your generative model for future projects?